

Contents

1	Appendix B: Environmental Channel Modeling	1
1.1	Objective	1
1.2	Planned Methods (src)	1
1.3	Planned Script and Outputs	1
1.4	Figure	1
1.5	Equation References	1
1.6	Reproducibility	1
1.7	Cross-References	1

1 Appendix B: Environmental Channel Modeling

1.1 Objective

Model atmospheric transmission beyond plateau windows with humidity/temperature/path effects; map channel capacity versus environment.

1.2 Planned Methods (src)

- `src/environmental_channel.py`
 - `atmospheric_transmission_detailed(wavelengths, humidity, temperature_k, path_m)`
 - `channel_capacity_vs_env(material_props, env_grid)`

1.3 Planned Script and Outputs

- Script: `scripts/generate_environmental_channel_analysis.py`
- Data: `output/data/environmental_channel.npz`
- Figure: `output/figures/environmental_channel_capacity.png`

1.4 Figure

1.5 Equation References

- Atmospheric transmission: see (??)
- Channel capacity: see (??)

1.6 Reproducibility

- Run: `python3 scripts/generate_environmental_channel_analysis.py`
- Artifacts saved to `output/data/` and `output/figures/`.

1.7 Cross-References

- Methods: ??
- Symbols: ??

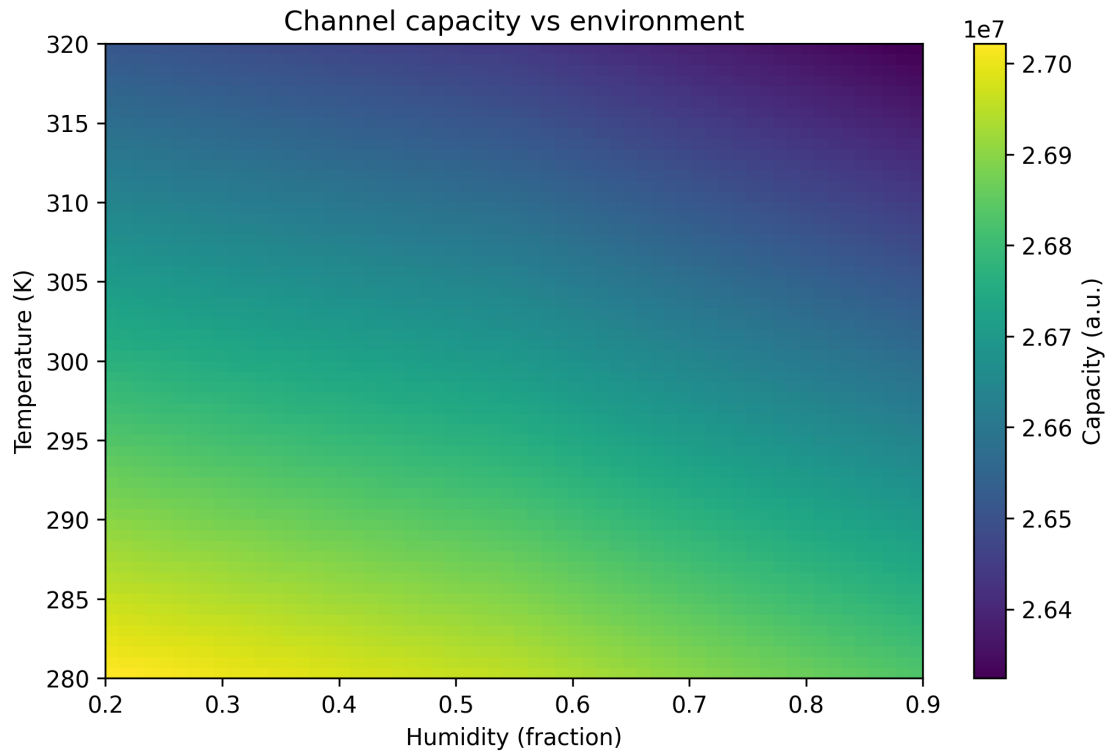


Figure 1: *Channel capacity as a function of humidity and temperature across IR wavelengths using detailed transmission modeling.*

- Math appendix: ??